1.


```
{
  "kind": "youtube#searchListResponse",
  "etag": "\"m2yskBQFythfE4irbTIeOgYYfBU/PaiEDiVxOyCWelLPuuwa9LKz3Gk\"",
  "nextPageToken": "CAUQAA",
  "regionCode": "KE",
  "pageInfo": {
    "totalResults": 4249,
    "resultsPerPage": 5
  },
  "items": [
    {
      "kind": "youtube#searchResult",
      "etag": "\"m2yskBQFythfE4irbTIeOgYYfBU/QpOIr3QKlV5EUlzfFcVvDiJT0hw\"",
      "id": {
        "kind": "youtube#channel",
        "channelId": "UCJowOS1R0FnhipXVqEnYU1A"
      }
    },
    {
      "kind": "youtube#searchResult",
      "etag": "\"m2yskBQFythfE4irbTIeOgYYfBU/AWutzVOt_5p1iLVifyBdfoSTf9E\"",
      "id": {
        "kind": "youtube#video",
        "videoId": "Eqa2nAAhHN0"
      }
    },
    {
      "kind": "youtube#searchResult",
      "etag": "\"m2yskBQFythfE4irbTIeOgYYfBU/2dIR9BTfr7QphpBuY3hPU-h5u-4\"",
      "id": {
        "kind": "youtube#video",
        "videoId": "lirngItQuVs"
      }
    }
  ]
}
```

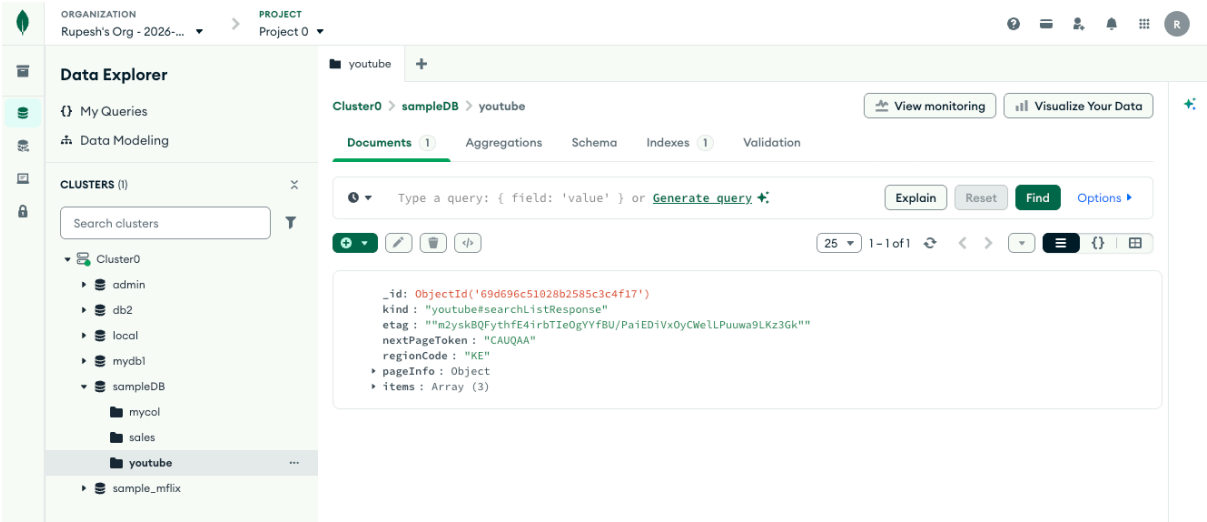

This is a sample data generated by youtube.

Expectations: I want you to provide commands

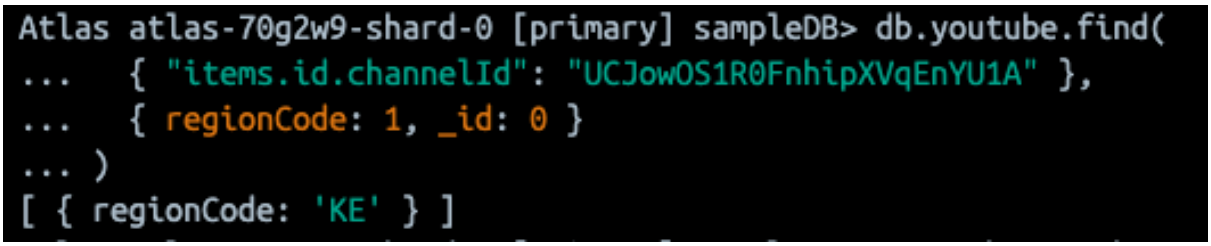
1. To insert this data into a mongo db collection with the name as **youtube**.

- First, we need to copy our mongosh command from ‘connect in cluster’ and then paste in our docker container to run shell through docker
- Command:
    - mongosh "mongodb+srv://cluster0.pao7yem.mongodb.net/" --apiVersion 1 --username admin

```
put password (Admin1234)
show dbs;
use sampleDB;
db.createCollection("youtube")
db.youtube.insert({
...   "kind": "youtube#searchListResponse",
...   "etag": "\"m2yskBQFythfE4irbTIeOgYYfBU/PaiEDiVxOyCWelLPuuwa9LKz3Gk\"",
...   "nextPageToken": "CAUQAA",
...   "regionCode": "KE",
...   "pageInfo": {
...     "totalResults": 4249,
...     "resultsPerPage": 5
...   },
...   "items": [
...     {
...       "kind": "youtube#searchResult",
...       "etag": "\"m2yskBQFythfE4irbTIeOgYYfBU/QpOIr3QKIV5EUlzfFcVvDiJT0hw\"",
...       "id": {
...         "kind": "youtube#channel",
...         "channelId": "UCJowOS1R0FnhipXVqEnYU1A"
...       }
...     },
...     {
...       "kind": "youtube#searchResult",
...       "etag": "\"m2yskBQFythfE4irbTIeOgYYfBU/AWutzVOt_5p1iLVifyBdfoSTf9E\"",
...       "id": {
...         "kind": "youtube#video",
...         "videoId": "Eqa2nAAhHN0"
...       }
...     },
...     {
...       "kind": "youtube#searchResult",
...       "etag": "\"m2yskBQFythfE4irbTIeOgYYfBU/2dIR9BTfr7QphpBuY3hPU-h5u-4\"",
...       "id": {
...         "kind": "youtube#video",
...         "videoId": "IirngItQuVs"
...       }
...     }
...   ]
... })
```



2. Mongodb command to return the only **regionCode** for only those docs where **channelId** is **UCJowOS1R0FnhipXVqEnYU1A**
   - db.youtube.find(
     {“items.id.channelId": "UCJowOS1R0FnhipXVqEnYU1A" },
     {regionCode: 1, _id: 0}
)



3. Write a pyspark code to read this document into spark dataframe.

   - First we need to exit out from mongo shell in docket using Ctrl + D
   - Then, we need to use connector script to connect between pyspark and mongodb
     - pyspark --packages org.mongodb.spark:mongo-spark-connector_2.12:3.0.1
   - Then, we need to use below command to read data from mongodb collections
     - from pyspark.sql.functions import explode, col
     - df = spark.read.format("com.mongodb.spark.sql.DefaultSource").option("uri", "mongodb+srv://admin:Admin1234@cluster0.pao7yem.mongodb.net/sampleDB.youtube").load()
     - df.show(truncate=False)
     - df.printSchema()

```
>>> df = spark.read.format("com.mongodb.spark.sql.DefaultSource").option("uri", "mongodb+srv://admin:Admin1234@c
luster0.pao7yem.mongodb.net/sampleDB.youtube").load()
>>> df.show(truncate=False)
+-------------------------+-------------------------------------------------------------+-------------------------
-------------------------------------------------------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------------
----------+-------------------------+-------------+---------+----------+
|_id                      |etag                                                         |items
                                                                                                                 |
          |kind                     |nextPageToken|pageInfo |regionCode|
+-------------------------+-------------------------------------------------------------+-------------------------
-------------------------------------------------------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------------
----------+-------------------------+-------------+---------+----------+
|{69d696c51028b2585c3c4f17}|"m2yskBQFythfE4irbTIeOgYYfBU/PaiEDiVxOyCWelLPuuwa9LKz3Gk"|[{"m2yskBQFythfE4irbTIeOgY
YfBU/QpOIr3QKlV5EUlzfFcVvDiJT0hw", {UCJowOS1R0FnhipXVqEnYU1A, youtube#channel, null}, youtube#searchResult}, {"m
2yskBQFythfE4irbTIeOgYYfBU/AWutzVOt_5p1iLVifyBdfoSTf9E", {null, youtube#video, Eqa2nAAhHN0}, youtube#searchResul
t}, {"m2yskBQFythfE4irbTIeOgYYfBU/2dIR9BTfr7QphpBuY3hPU-h5u-4", {null, youtube#video, IirngItQuVs}, youtube#sear
chResult}]|youtube#searchListResponse|CAUQAA       |{4249, 5}|KE        |
+-------------------------+-------------------------------------------------------------+-------------------------
-------------------------------------------------------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------------------
----------+-------------------------+-------------+---------+----------+

>>> df.printSchema()
root
 |-- _id: struct (nullable = true)
 |    |-- oid: string (nullable = true)
 |-- etag: string (nullable = true)
 |-- items: array (nullable = true)
 |    |-- element: struct (containsNull = true)
 |    |    |-- etag: string (nullable = true)
 |    |    |-- id: struct (nullable = true)
 |    |    |    |-- channelId: string (nullable = true)
 |    |    |    |-- kind: string (nullable = true)
 |    |    |    |-- videoId: string (nullable = true)
 |    |    |-- kind: string (nullable = true)
 |-- kind: string (nullable = true)
 |-- nextPageToken: string (nullable = true)
 |-- pageInfo: struct (nullable = true)
 |    |-- totalResults: integer (nullable = true)
 |    |-- resultsPerPage: integer (nullable = true)
 |-- regionCode: string (nullable = true)
```

4. Pyspark code to return the only **regionCode** for only those docs where **channelId** is **UCJowOS1R0FnhipXVqEnYU1A**

- For this question, I created a dataframe that explodes the column ‘items’
- Then I filtered out exploded column ‘items’ using below script
    - df_exploded = df.withColumn("item", explode(col("items")))
    - result = df_exploded.filter(col("item.id.channelId") == "UCJowOS1R0FnhipXVqEnYU1A").select("regionCode")
    - result.show()

```
>>> df_exploded = df.withColumn("item", explode(col("items")))
>>> result = df_exploded.filter(col("item.id.channelId") == "UCJowOS1R0FnhipXVqEnYU1A").select("regionCode")
>>> result.show()
+----------+
|regionCode|
+----------+
|        KE|
+----------+
```

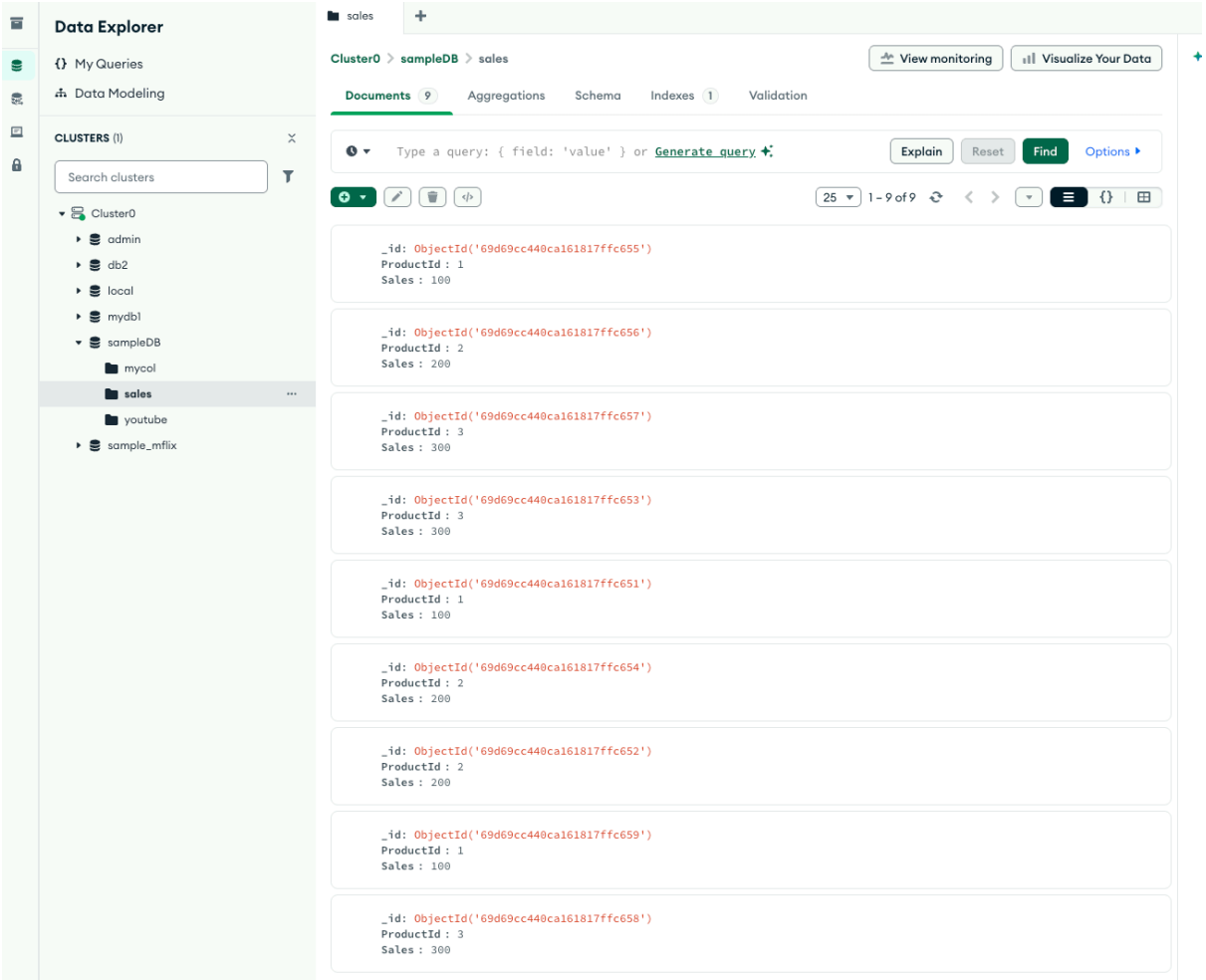
sales.csv

```
ProductId, Sales
1,100
2,200
3,300
1,100
2,200
3,300
1,100
2,200
3,300
```

Q. Create sales.csv in spark dafarame and write to mongodb in sales collection.

```
>>> sales_data = [
...     (1, 100),
...     (2, 200),
...     (3, 300),
...     (1, 100),
...     (2, 200),
...     (3, 300),
...     (1, 100),
...     (2, 200),
...     (3, 300)
... ]
>>>
>>> sales_df = spark.createDataFrame(sales_data, ["ProductId", "Sales"])
>>> sales_df.show()
+---------+-----+
|ProductId|Sales|
+---------+-----+
|        1|  100|
|        2|  200|
|        3|  300|
|        1|  100|
|        2|  200|
|        3|  300|
|        1|  100|
|        2|  200|
|        3|  300|
+---------+-----+

>>> sales_df.write.format("com.mongodb.spark.sql.DefaultSource").mode("append").option("uri", "mongodb+srv://adm
in:Admin1234@cluster0.pao7yem.mongodb.net/sampleDB.sales").save()
```


Data Explorer
My Queries
Data Modeling
CLUSTERS (1)
Search clusters
Cluster0
admin
db2
local
mydb1
sampleDB
mycol
sales
youtube
sample_mflix
sales
Cluster0 > sampleDB > sales
View monitoring
Visualize Your Data
Documents 9
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
Options
25
1 - 9 of 9
_id: ObjectId('69d69cc440ca161817ffc655')
ProductId : 1
Sales : 100
_id: ObjectId('69d69cc440ca161817ffc656')
ProductId : 2
Sales : 200
_id: ObjectId('69d69cc440ca161817ffc657')
ProductId : 3
Sales : 300
_id: ObjectId('69d69cc440ca161817ffc653')
ProductId : 3
Sales : 300
_id: ObjectId('69d69cc440ca161817ffc651')
ProductId : 1
Sales : 100
_id: ObjectId('69d69cc440ca161817ffc654')
ProductId : 2
Sales : 200
_id: ObjectId('69d69cc440ca161817ffc652')
ProductId : 2
Sales : 200
_id: ObjectId('69d69cc440ca161817ffc659')
ProductId : 1
Sales : 100
_id: ObjectId('69d69cc440ca161817ffc658')
ProductId : 3
Sales : 300